Lucas Zhang

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Education

BASIS Peoria Peoria, AZ

High School Diploma Candidate *Expected Graduation: May 2027*

- GPA: 3.91 Unweighted / 4.71 Weighted; 16 AP courses including AP Computer Science A, AP Computer Science Principles, AP Calculus BC, AP Physics 1/2, AP Chemistry, AP Research, and AP Seminar

Technical Skills

Languages: C, C++, Python, Java ,TypeScript , SQL, HTML/CSS/JS, Assembly
Frameworks & Libraries: React, Node.js, Express, FastAPI, NumPy, pandas, Git, Docker, Ghidra, IDA Pro, x64dbg
AI / Robotics: PyTorch, TensorFlow, YOLO, OpenCV, ROS 2, Nav2, SLAM, Arduino

Projects

GenesisOS (CustomOS) | *C, Assembly, Linux tooling* | GitHub *Solo*

- Built a from-scratch higher-half x86_64 operating system spanning PMM/VMM, preemptive scheduling, ELF userland, VFS/FAT32, TCP/IP, USB (xHCI), HD Audio, and a software-rendered desktop/window manager
- Validated subsystem correctness with a consolidated `MASTER_SELFTEST` harness, achieving 26/26 PASS on QEMU `q35`

mycc | *C++, LLVM, x86-64* | GitHub *Solo*

- Engineered a from-scratch C subset compiler with hand-written lexer/parser/semantics and dual backends (handwritten x86-64 assembly + LLVM IR)
- Verified native executable correctness against GCC using a 117-program differential suite across 12 build stages, with identical results on both backends

GridMark | *ROS 2, Nav2, YOLO, Python, C++* | GitHub *Solo*

- Designed an autonomous indoor SLAM robot that maps with LiDAR, explores via Nav2 frontiers, and detects targets with fine-tuned YOLO on Jetson Orin Nano Super
- Integrated Jetson compute with Arduino Mega motor/encoder control over USB serial, including fail-safe command timeout and calibrated odometry/TF geometry

AZSEF Water Analysis | *Python, TypeScript, React, PostgreSQL* | GitHub *Solo*

- Developed five independent Gaussian Process regression models to predict dissolved Fe/Cr/Mn/Mo/In across Phoenix from six FAAS-analyzed field samples
- Shipped a React web app with serverless APIs for click-to-predict maps with calibrated posterior uncertainty and LOOCV validation; recognized with the Stockholm Junior Water Prize at AzSEF

CPU Path Tracer | *C, Multithreading, Algorithms* | GitHub *Solo*

- Implemented a dependency-free multithreaded path tracer with Lambertian/metal/dielectric materials, Monte Carlo GI, BVH acceleration, and deterministic multi-core rendering

Experience

National Underwater Robotics Challenge (NURC) / ASU Underwater Robotics Phoenix, AZ

Lead Programmer (Team ∼17) *2026*

- Led software development for tethered underwater ROV platforms, coordinating controls, mission logic, and integration across a ∼17-person engineering team
- Directed programming for pool testing and competition runs; team placed 3rd at the National Underwater Robotics Challenge

ASU EPICS - BMECS (Tiger Mountain Foundation) Phoenix, AZ

Student Engineer (Team of 4) *2026*

- Engineered a volunteer-operable bed-level environmental control kit to reduce garden heat stress and water waste without buried irrigation infrastructure
- Presented project work at Arizona State University Ira A. Fulton Schools of Engineering

Leadership & Activities

Speech & Debate BASIS Peoria

High School Captain; Debate BQ D2 State Champion

- Led a high school team of ∼100-120 students as Captain after captaining a middle school team of ∼50; won Debate BQ D2 State Championship

Olympiad & Competitive Programming Independent / Contests

Codeforces Master; USACO Gold

- Train and compete in olympiad-style algorithmic contests; reached Codeforces Master (peak rating 2129) and USACO Gold through sustained problem-solving practice

STEM Clubs BASIS Peoria

Member

- Active in Robotics Club, Coding Club, Math Club, Circuits Club, Electrical Engineering Club, and Machine Learning Club

National Chinese Honor Society BASIS Peoria

President

- Directed chapter programming and membership initiatives; supported Chinese culture education volunteering